Can a methane detector see carbon monoxide?

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The short version

Satellites can see plumes of gas from space. A group at SRON in the Netherlands built a machine-learning system that automatically finds plumes of methane in satellite images. I asked a simple question, if that system was only ever taught to recognise shapes, does it also work on a completely different gas - carbon monoxide?

The answer is neither yes nor no, and the third answer is the useful one.

> **What we found**
>
> The shape-recognition transfers. A methane-trained network picks Indian steel plants out of their own regional background, and shows detected CO plume.
>
> When a scene had many cloud gaps, the score dropped. That effect turned out to be about three times stronger than the effect of a steel plant actually being there.

1. What a satellite actually sees

A satellite called Sentinel-5P circles the Earth roughly fourteen times a day and measures how much radiation comes back. Different gases absorb at different wavelengths, so from the absorption lines in received radiation, how much of a given gas is present in the column of air between the ground and space.

It does this for a patch of ground (spatial resolution), so each pixel holds a gas amount rather than a brightness.

From the emission sources, the wind smears the gases downwind into a shape, wider and fainter as it travels. From above it looks a bit like steam drifting off a kettle, seen from the ceiling. That characteristic smear is what a plume detector is looking for.

Carbon monoxide is produced whenever something burns incompletely. Steelmaking is unusually good at making it: blast furnaces deliberately use carbon monoxide to strip oxygen out of iron ore, and some always escapes. A steel plant emits roughly six times more carbon monoxide per unit of carbon dioxide than the industrial average. Carbon monoxide has a much lower background, so a steel plant is a prominent source. In an urban region, there are many contributing sources, and emission hotspots are observable.

Why reuse a methane model?

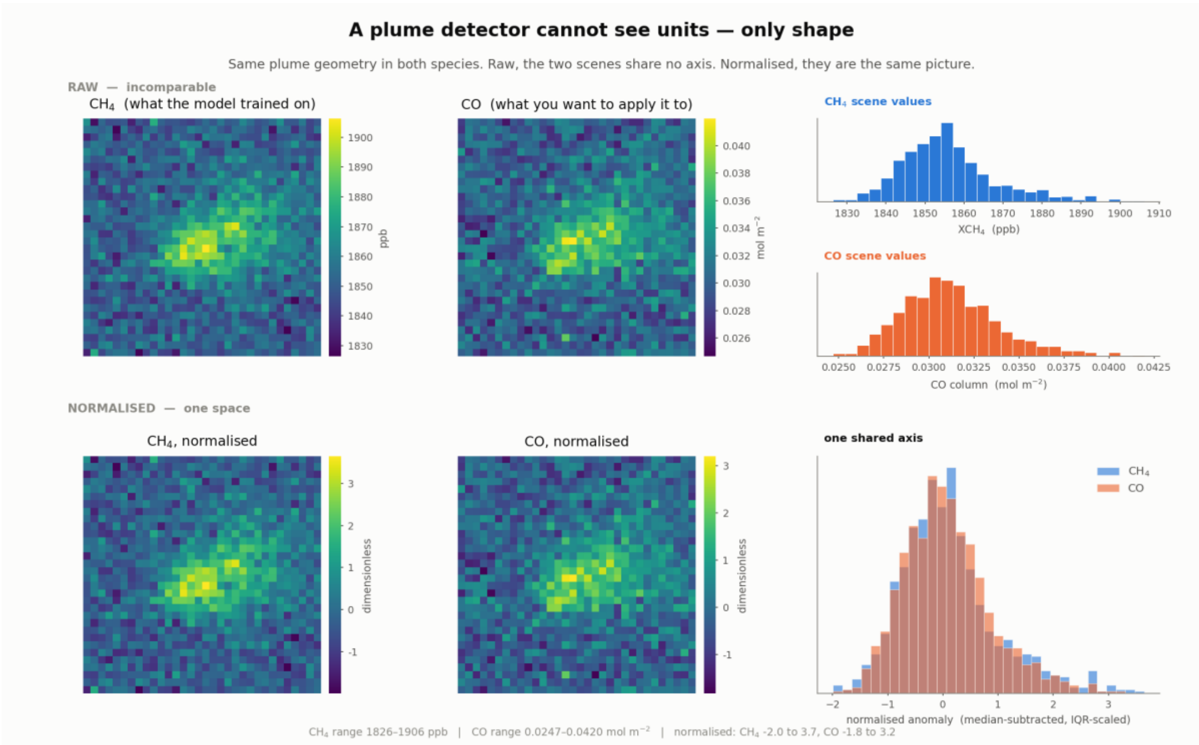
Teaching a computer to recognise plumes needs thousands of labelled examples, and someone has to look at every one of them. That work has already been done for methane. It has not been done for carbon monoxide.

But it could be treated as if the methane model was never told it was looking at methane. It was shown small square images and asked whether the pattern looked plume-like. If those images are stripped of their units first, the model genuinely cannot know which gas it is reading. So it might just work on another one.

2. How swap is possible?

Methane and carbon monoxide have different concentrations in the air, and vary differently from day to day. Feeding raw carbon monoxide numbers to a model trained on methane numbers would be like handing a thermometer reading to someone expecting a bank balance.

So we remove the units before the model ever sees the image. For each scene, we find the middle value, subtract it, and divide by how spread out the rest of the values are. What comes out is not “parts per billion of anything”, it is simply how unusual each pixel is compared with its neighbours.



Top row: the same plume as methane and as carbon monoxide, raw - they cannot even share an axis. Bottom row: after normalising, they are statistically the same picture.

The same plume is a 2.8% bump above background in methane, and a 35.5% bump in carbon monoxide. After normalising, both peak at about 3.2 to 3.7.

> **The asymmetry, and why it does not mean CO is easier**
>
> A 35% bump sounds far easier to spot than a 3% one. It is not as carbon monoxide's background moves around much more from day to day, and fires could put plume-shaped carbon monoxide everywhere. The difficulty moves from finding the plume to deciding what it is.

3. What we trained, and what we collected

The model is small by modern standards, about 590,000 adjustable numbers, arranged as two pattern-finding layers followed by two decision layers. It reads a 32 by 32 pixel square, roughly 220 kilometres on a side and returns a single number between 0 and 1: how plume-like does this look?

Before trying anything new, we checked the model on methane — the gas it was built for.

Out of every 100 real plumes, it found 97.

Out of every 100 things it flagged, 96 were genuine plumes.

These scores match the ones reported in the original paper. That was the point of the check: it told us our rebuilt copy of the model was working properly.

Then, without retraining a single thing, we pointed it at carbon monoxide over 26 places in four groups:

Group	Sites	Why
Indian steel plants	13	Large, poorly monitored, no public facility reporting
European steel plants	7	Emissions already published — the honest test
Indian cities	3	Big but diffuse: traffic, cooking, industry
Empty places	3	Sahara, open Pacific, a clean Himalayan site

That last group matters most. If a detector lights up over the middle of the Sahara, it is finding faults in the data, not gas in the air.

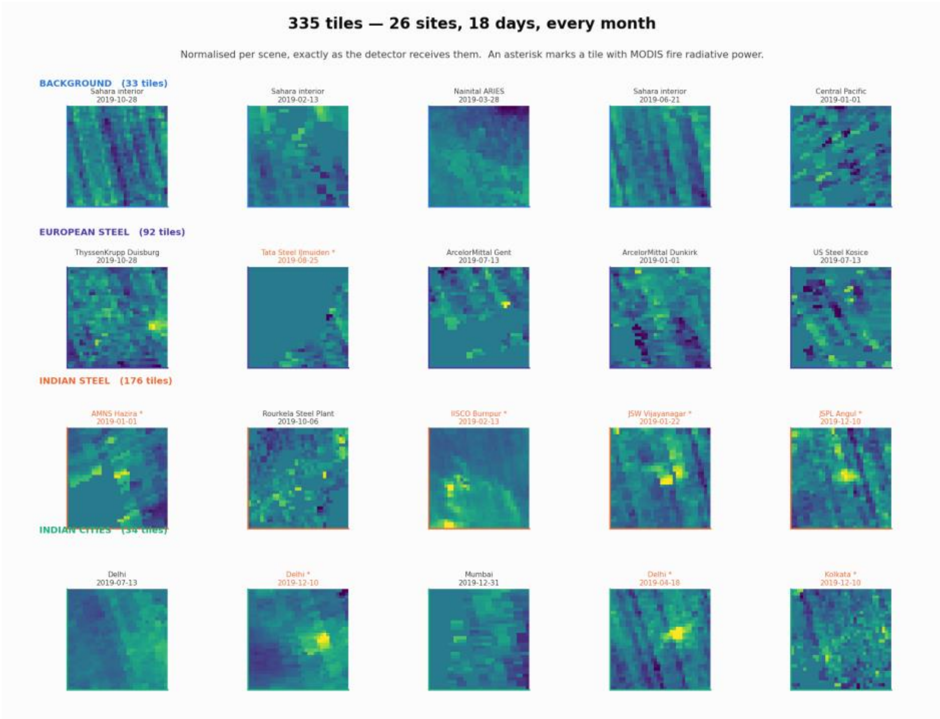
What we actually got

Quantity	Value
Scenes collected	335, from 26 sites on 18 days across 2019
Site-days attempted	468 — so 28% were dropped, mostly to cloud
Usable pixels per scene	81% on average
CO range	0.0222 to 0.0590 mol m^{-2} (1st to 99th percenscene)
Nitrogen dioxide available	333 of 335 scenes
MODIS fire signal present	128 of 335 scenes

Every site was sampled on the same 18 days, so no region got only its clear season. Hold on to the 28% figure — it comes back to matter.

4. What 335 scenes look like

Before any statistics, it is worth simply looking at the pictures the detector is being handed.



A sample of the 335 scenes, normalised after filteration. Rows are the four groups; an asterisk marks a scene where MODIS saw a fire.

Two textures keep recurring, and both matter.

- Compact bright cores near the middle, like a hotspot, sometimes with a tail-like plume.
- Regular diagonal banding running corner to corner. This is not atmosphere, rather it is an instrument artefact. The satellite scans in strips, and small differences between detector columns leave stripes; the re-gridding turns them diagonal.

The striping is the awkward one. A stripe is straight, elongated and coherent, which is also a fair description of a plume. This is exactly the class of false alarm that the published method needs a whole second stage to reject.

5. A sanity check that passed

Before asking whether the detector works, ask whether the data is real. If the underlying carbon monoxide amounts do not line up with what we already know about the world, nothing built on top of them will mean anything.

Group	Median CO (mol m^{-2})	Reading
Remote	0.0290	Clean marine and desert air
European steel	0.0321	Moderately polluted continental air
Indian cities	0.0401	Indo-Gangetic Plain and coastal
Indian steel	0.0409	Same, plus local industry

Nothing here is surprising, and that is precisely the point — it tells you the pipeline is sound. Clean air at the bottom, the winter Indo-Gangetic Plain at the top.

6. The ranking, site by site

Here is the raw output — every site, ranked by its average score across the year. This is the table that looks like a result, and it is the one to be most careful with.

#	Site	Mean plume score	Scenes	Usable data
1	**IISCO Burnpur**	0.840	15	94%
2	Bokaro Steel Plant	0.779	17	91%
3	Bhilai Steel Plant	0.735	14	90%
4	Durgapur Steel Plant	0.718	15	94%
5	Rourkela Steel Plant	0.695	16	91%
6	Tata Steel Jamshedpur	0.659	15	92%
7	Tata Steel Kalinganagar	0.561	14	92%
8	JSPL Angul	0.514	15	93%
9	JSPL Raigarh	0.461	15	91%
10	Kolkata	0.369	15	88%
11	Delhi	0.357	13	83%
12	**Nainital ARIES**	0.300	15	84%
13	**Central Pacific**	0.296	3	79%
14	**ThyssenKrupp Duisburg**	0.276	12	76%
15	JSW Vijayanagar	0.257	12	81%
16	US Steel Kosice	0.246	17	74%
17	Vizag Steel RINL	0.174	12	82%
18	ArcelorMittal Gent	0.168	13	69%
19	AMNS Hazira	0.144	8	63%
20	ArcelorMittal Gijon	0.108	11	70%
21	ArcelorMittal Taranto	0.061	13	67%
22	ArcelorMittal Dunkirk	0.056	14	54%
23	Tata Steel IJmuiden	0.055	12	61%
24	Sahara interior	0.030	15	82%
25	**JSW Dolvi**	0.011	8	66%
26	**Mumbai**	0.006	6	69%

All 26 sites ranked by mean plume score. The right-hand column is the share of each scene that contained real data.

The top of the list is encouraging: eight of the top nine are Indian steel plants, with the two big Indian cities just below them. Read no further, and you would conclude the detector works well.

But four entries should stop you.

- IISCO Burnpur is first, at 0.840. It makes about 2.5 million tonnes of steel a year — roughly a quarter of Tata Jamshedpur, which is sixth. It also sits in the middle of the Durgapur–Asansol industrial belt, 30 km from another steel plant on the list, surrounded by coal power stations.
- Nainital and the Central Pacific — two of our deliberately empty control sites — score 0.300 and 0.296, above ThyssenKrupp Duisburg at 0.276. Duisburg is the largest steelworks in Europe.

- JSW Dolvi is 25th out of 26, at 0.011. It produces about ten million tonnes of steel a year. Mumbai, a city of twenty million people, is last.
- The Sahara, which should be dead last, is 24th, above two working steel plants.

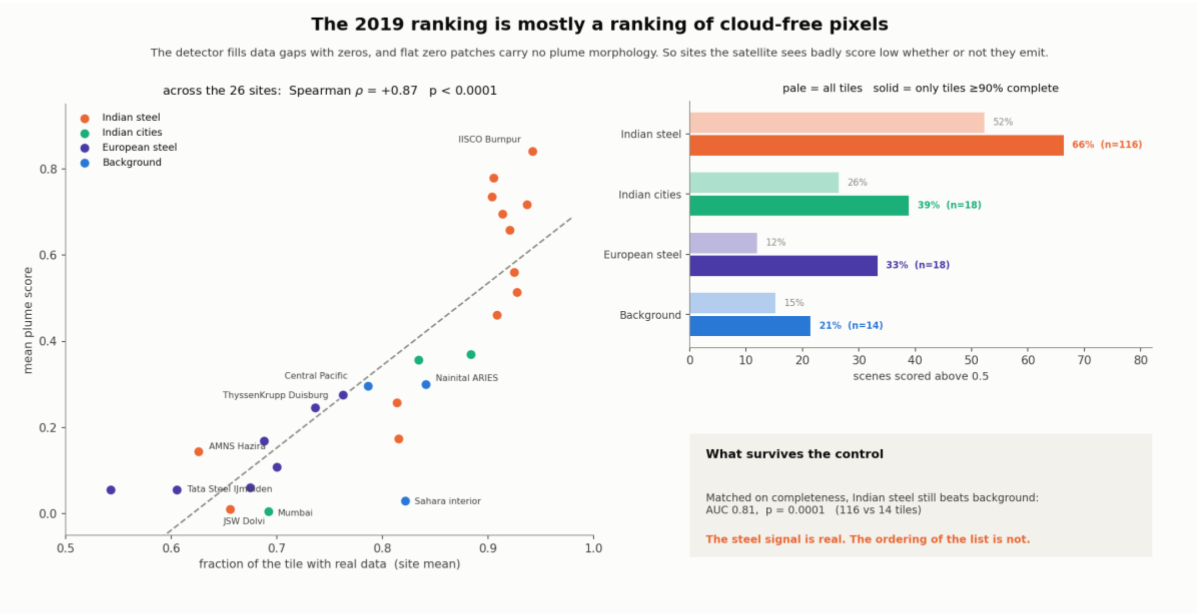
None of that is believable as a statement about emissions. So the question becomes: what else could these scores be tracking?

7. The scores are partly grading the weather

Look again at the last column of that table. The top of the list runs at 90-94% usable data. The bottom runs at 54-70%.

> **The mechanism**
>
> Satellites cannot see through cloud, and the carbon monoxide measurement needs sunlight reflected off a bright surface. Where it fails, the image has a hole/not valid pixel in it.
>
> Our code fills those holes with zeros before showing the image to the model.
>
> A patch of zeros is perfectly flat, and a flat patch contains no plume shape at all.
>
> So an image with many holes is pushed towards a low score no matter what is underneath it.

The size of this effect is much larger than expected. Across the 26 sites, average score and fraction of usable data move together with a correlation of 0.87, close to a straight line.

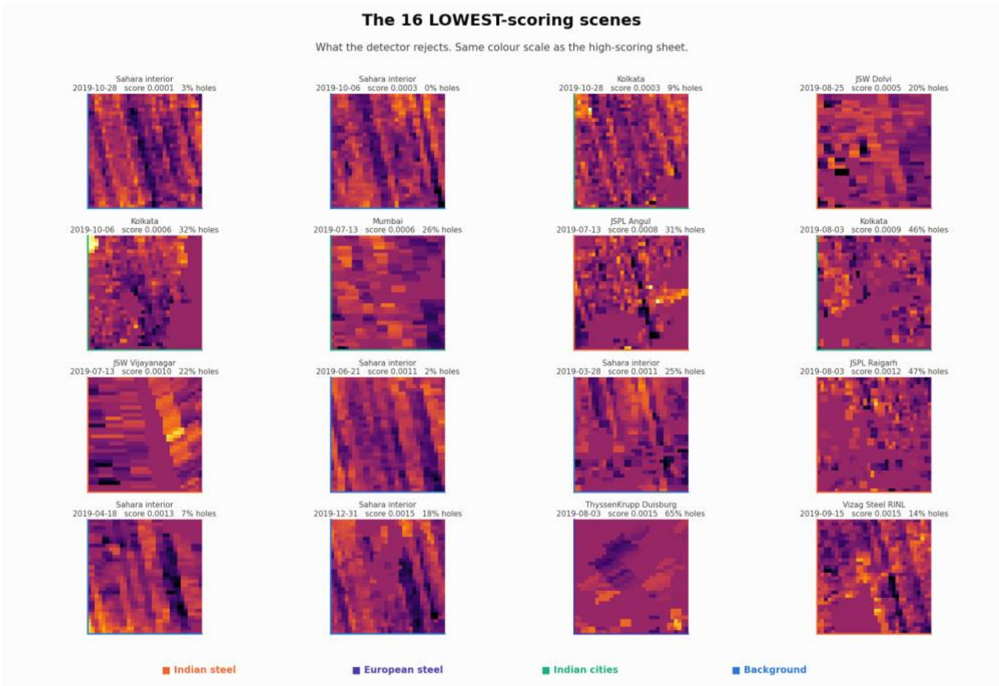


Left: each dot is one site. Almost all the spread in the ranking is explained by how much usable data the satellite got. Righ t: what happens when you compare only the clearest images.

The three sites at the bottom are JSW Dolvi, Mumbai, AMNS Hazira, are all coastal. The sea is dark at the wavelengths this measurement uses, so the retrieval mostly fails over water. Half of each of those images was missing. Their low scores say nothing at all about their emissions.

8. What the detector throws away

Anyone can show the cases where a method works. What it rejects tells you what it has actually learned, so here are the sixteen lowest-scoring scenes in the whole set.



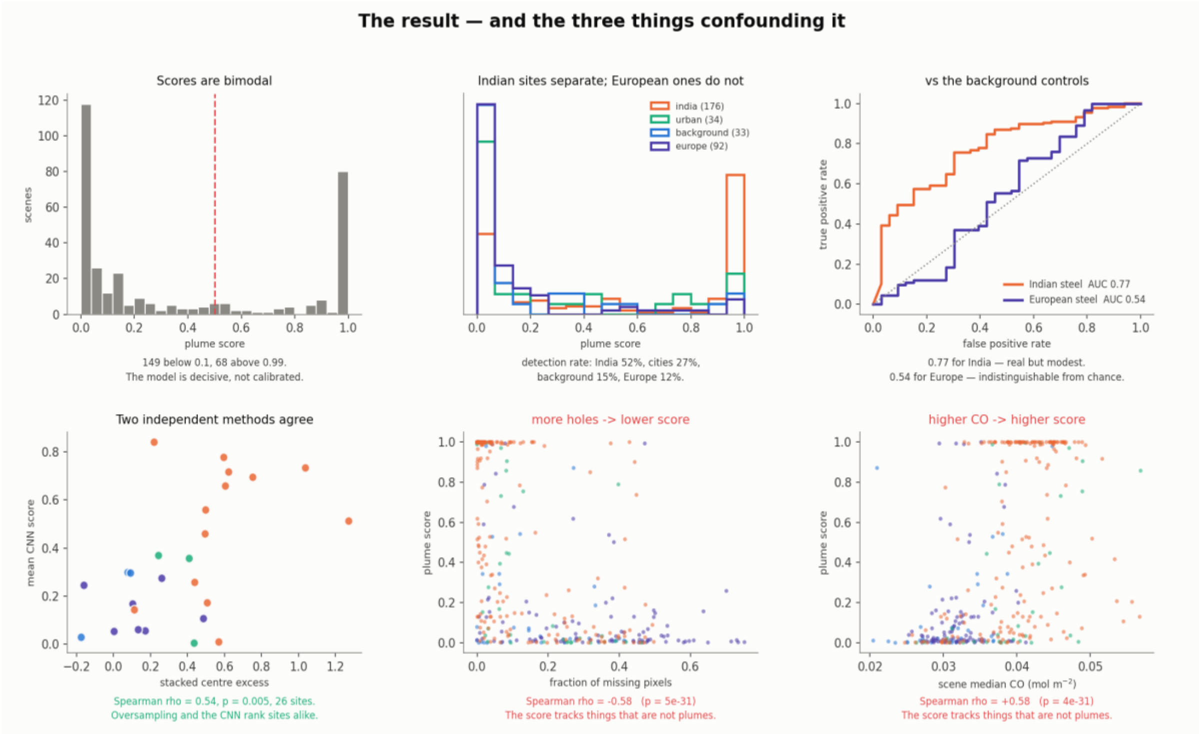
The 16 lowest-scoring scenes, on the same colour scale as the high scorers.

Almost everyone is dominated by the diagonal banding described earlier, the Sahara appearing at scores of 0.0001, 0.0003, and 0.0011. This is a genuinely positive aspect, and it corrected a prediction we had made in the other direction, the worry was that the detector would latch onto instrument stripes and report them as plumes. It does the opposite. Regular periodic banding evidently does not look like a plume to it.

The second pattern here is less welcome. Several of the low scorers have large missing-data fractions - 65%, 47%, 46%. That is the confound from the previous section, showing up in the pictures.

9. Does the signal survive the confounds?

The obvious worry is that the detector has simply learned to recognise “India”. Three tests say otherwise.

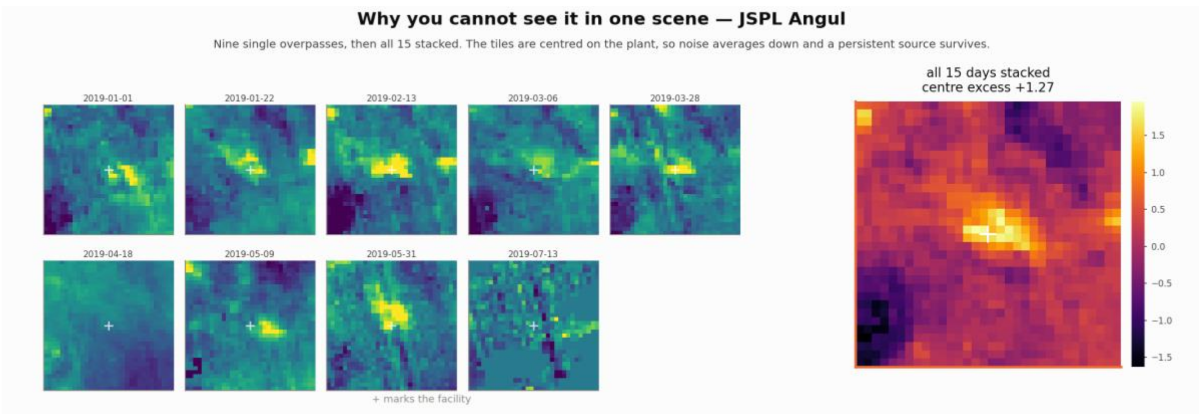


Top row: the scores are strongly bimodal; Indian sites separate from controls while European ones do not; the accuracy curves. Bottom row: the three things confounding the result.

A few readings from that figure. The model is decisive rather than calibrated, 149 scenes score below 0.1 and 68 above 0.99, so a score of 0.99 is not a 99% probability, and moving the threshold barely changes anything. Indian steel reaches an accuracy of 0.77 against the control sites using all scenes; European steel reaches 0.54, which is a chance.

10. One day is never enough

At 7-kilometre resolution, one overpass usually may not show a steel-plant plume at all. So we do what astronomers do with faint stars, leave the shutter open. Every scene is centred on the same facility, so averaging a site's days is exactly the oversampling technique the literature uses. Random noise averages down; anything permanently there survives.



Nine ordinary days at JSPL Angul, then all fifteen averaged. The blob only appears in the average.

11. Stack scenes and then plume detection

It is observed that stacking improves the scene and may lead to improved plume detection. So the images are stacked over each individual site, but since it is observed in section 7 that if we consider the scenes in which certain valid pixels are present, then the plume score increases. This is like rejecting those scenes which are corrupted with clouds or retrieval uncertainties. So to have a better idea, we have stacked under 3 cases: (a) no filter, all images stacked, (b) 66.6% valid pixels per scene, (c) 90% valid scenes.

For example, if stacking is happening over Kolkata and one scene/image has fewer than 66.6% valid pixels, then that scene/image will not be used in the stacked image for plume detection.

How many scenes does each filter keep

group	No. of scenes with no filter	No. of scenes with ≥ 66.6% valid pixel	No. of scenes with ≥ 90% valid pixel
Indian steel	176	150 (85%)	116 (66%)
Indian cities	34	28 (82%)	18 (53%)
remote controls	33	27 (82%)	14 (42%)
European steel	92	44 (48%)	**18 (20%)**
all scenes	335	249 (74%)	166 (50%)

Europe keeps one scene in five at the 90% cut. Five sites are left with nothing at all - ArcelorMittal Dunkirk, ArcelorMittal Taranto, JSW Dolvi, AMNS Hazira and Mumbai - and Central Pacific keeps a single scene.

Mean plume score by category

category	Mean plume score with no filter	Mean plume score with ≥ 66.6% valid pixel	Mean plume score with ≥ 66.6% valid pixel
India steel	0.716 (13)	0.739 (13)	0.844 (11)
India cities	0.160 (3)	0.186 (3)	0.414 (2)
EU steel	0.380 (7)	0.292 (7)	0.309 (5)
background	0.476 (3)	0.459 (3)	0.331 (3)

Mean stacked score across the sites in each category, with the number of sites still scoreable in brackets.

Stacked plume score, site by site

site	group	n	Plume score with no filter	n	Plume score ≥ 66.6% valid pixels	n	Score ≥ 90% valid pixels
Bhilai Steel Plant	India steel	14	1.000	12	1.000	10	1.000
Tata Steel Jamshedpur	India steel	15	1.000	13	1.000	12	1.000
JSPL Angul	India steel	15	1.000	14	1.000	11	1.000
Tata Steel Kalinganagar	India steel	14	1.000	13	1.000	13	1.000
Rourkela Steel Plant	India steel	16	0.999	15	1.000	13	1.000

site	group	n	Plume score with no filter	n	Plume score ≥ 66.6% valid pixels	n	Score ≥ 90% valid pixels
Durgapur Steel Plant	India steel	15	0.975	15	0.975	12	0.999
JSPL Raigarh	India steel	15	0.758	14	0.938	11	0.990
IISCO Burnpur	India steel	15	0.987	15	0.987	11	0.976
Bokaro Steel Plant	India steel	17	0.925	15	0.904	12	0.805
JSW Vijayanagar	India steel	12	0.298	10	0.273	5	0.355
Vizag Steel RINL	India steel	12	0.189	8	0.223	6	0.158
AMNS Hazira	India steel	8	0.141	3	0.276	0	—
JSW Dolvi	India steel	8	0.038	3	0.035	0	—
US Steel Košice	EU steel	17	0.703	12	0.543	3	0.783
ThyssenKrupp Duisburg	EU steel	12	0.010	9	0.130	6	0.504
ArcelorMittal Gijón	EU steel	11	0.103	4	0.148	3	0.179
ArcelorMittal Gent	EU steel	13	0.136	5	0.058	3	0.070
Tata Steel IJmuiden	EU steel	12	0.085	5	0.082	3	0.009
ArcelorMittal Taranto	EU steel	13	0.943	6	0.952	0	—
ArcelorMittal Dunkirk	EU steel	14	0.679	3	0.132	0	—
Kolkata	India cities	15	0.081	14	0.396	11	0.631
Delhi	India cities	13	0.240	10	0.112	7	0.197
Mumbai	India cities	6	0.159	4	0.048	0	—
Nainital ARIES	background	15	0.439	13	0.385	6	0.983
Central Pacific	background	3	0.988	3	0.988	1	0.007
Sahara interior	background	15	0.002	11	0.005	7	0.001

Scene count and stacked plume score at each cut: all images with no filter, then ≥ 66.6% complete, then ≥ 90%. Sorted by the score on the clearest scenes. n is the number of scenes.

Indian steel rises steadily and modestly

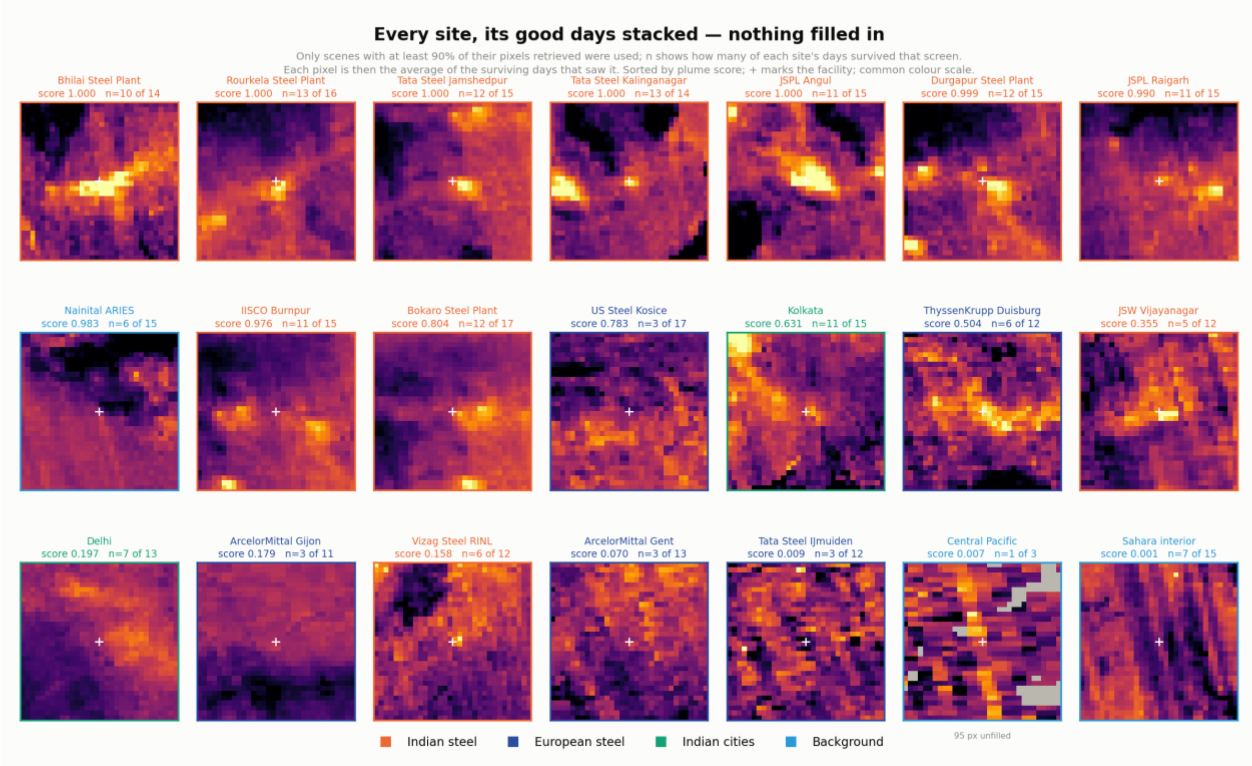
0.716 to 0.739 to 0.844. The plants that were already at 1.000 stay there; the gain comes from the middle of the list. JSPL Raigarh moves 0.758 → 0.938 → 0.990 and Durgapur 0.975 → 0.999. Nothing in this group changes character as it simply sharpens.

Two plants go the other way. Bokaro falls 0.925 → 0.805 and IISCO Burnpur 0.987 → 0.976, both having lost a third of their scenes.

Europe is where the interesting changes are

ThyssenKrupp Duisburg goes 0.010 → 0.130 → 0.504. On all its images, it is the lowest-scoring European site in the set; on its six clearest (>90% valid pixel), it crosses the detection threshold. That is the cloud confound lifting, Duisburg's scenes were being dragged down by their holes, not by an absence of emissions.

But the category mean does not follow, because two of the European numbers on the left were never real. ArcelorMittal Taranto scores 0.943 on all images and 0.952 at 66.6%, and ArcelorMittal Dunkirk 0.679 — both built on composites where a large share of pixels rested on one or two observations. At the 90% pixel filter, neither site has a single scene, and the artefacts go with them. The European mean falls from 0.380 to 0.309 while the group's one site, ThyssenKrupp Duisburg, rises.



Stacked scene with 90 % valid pixels for each scene, sorted by plume score. 'n = 10 of 14' means ten of the site's fourteen days passed. The five sites that lost every scene are absent. Grey at Central Pacific marks pixels its single surviving day never saw.

The controls diverge at remote sites

Central Pacific sits at 0.988 on all images and at 66.6%, a remote ocean site with no source, scoring higher than eight steel plants. It has three scenes, and half its composite rests on one or two observations. At the 90% pixel filter, it keeps one scene and falls to 0.007, which is what it should have read all along.

The Sahara behaves throughout: 0.002, 0.005, 0.001. Nainital ARIES does not. It moves 0.439 → 0.385 → **0.983**, above eight of the eleven surviving Indian plants. Its six clearest scenes score 1.00, 0.98, 0.53, 0.34, 0.34 and 0.29 individually; the two clearest days are the two that look most like a source. Nainital overlooks the Indo-Gangetic Plain, so it could be due to haze as well.

So the 90% cut removes one false positive and creates another. It is not a straightforward improvement to the control group; it changes which control misbehaves.

12. What holds up, and what does not

Holds up:

- A methane-trained network, with no retraining, detects carbon monoxide plumes over large Indian steel plants.
- The same plants score 59% in the dry season against 21% in the monsoon, depicting lossing important information with high cloud coverage, while remote sites show no swing at all.
- Stacking a site's scenes brings a low signal into a visible one. On the unfiltered stack, Indian steel reaches +0.59 centre excess while the remote sites sit at −0.003.
- Rejecting the cloudiest scenes before stacking sharpens it further. Indian steel rises from a mean plume score of 0.716 with no filter to 0.844 at the 90% filter, and ThyssenKrupp Duisburg from 0.010 to 0.504, a plant that is undetected on all its images and detected on its six clearest.
- The filter (>90%) removes two scores that were never real. ArcelorMittal Taranto at 0.943 and Central Pacific at 0.988 were both built on composites where a large share of pixels rested on one or two observations, and neither survives the 90% filter. That these disappear while Duisburg appears is the clearest evidence that the filter is doing physical work rather than simply discarding data.
- The detector rejects instrument stripes rather than counting on them.
- The regional carbon monoxide gradient comes out exactly as physics predicts.

Does not hold up:

- The ranking of individual facilities. It mostly reflects cloud cover and coastline, not emissions, and should not be quoted. Filtering improves it for sites that retain ten or more scenes, but does not make it safe to quote for the rest.
- Using a filter leads to a loss of scenes. Half of all scenes fail the 90% filter, European steel keeps one scene in five, and five sites are left with nothing at all. The scene count belongs beside every score.

Sources

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Sentinel-5P TROPOMI carbon monoxide and nitrogen dioxide, and MODIS fire radiative power, accessed through Google Earth Engine. Training images from Schuit et al., Zenodo record 13903869, CC-BY 4.0.